

Daren Xu

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EDUCATION

Northeastern University, Boston, MA

GPA: 3.8 / 4.0 | Dean's List (Spring 2026)

Candidate for Bachelor of Science in Electrical Engineering

Expected May 2028

Relevant Courses: Fundamentals of Electronics & Lab, Fundamentals of Linear Systems, Analysis of Random Phenomena in ECE, Manufacturing Systems & Techniques, Embedded Design: Enabling Robotics (FPGA), Circuits and Signals: Biomedical Applications

EXPERIENCE

Hangzhou Boomy Intelligent Technology Co., Ltd.

Hangzhou, China | August 2026

Robotics R&D Intern — Real-Time Audio / DSP

- Designed the robot-side full-duplex acoustic echo canceller for a robot-to-operator intercom on a Rockchip RK3588S SoC: a multichannel frequency-domain Kalman filter with Hammerstein saturation modeling of loudspeaker nonlinearity, Wiener residual suppression, and minimum-statistics noise reduction — 3,700 lines of Python (numpy/scipy) across 16 modules.
- Built a 15-scene acoustic simulator and a gaming-resistant metric suite after showing that the standard ERLE metric rewards half-duplex microphone gating; the final configuration averages 26.8 dB ERLE and 15.3 dB noise attenuation across the suite at 21 ms latency and 9.9% of one CPU core.
- Diagnosed loudspeaker distortion as saturation, not polynomial, after measuring 70,000x crest-factor amplification and 1e9+ condition numbers on speech; replaced the diverging polynomial basis with a bounded tanh basis, ending a failure mode where nonlinear branches absorbed 95% of filter energy.
- Eliminated clock drift architecturally by running playback and capture in one duplex callback on a single sound card; brought up and calibrated the audio path on two SoC platforms (Raspberry Pi 4B, LubanCat-4) and traced a 6 dB performance ceiling to board supply noise, cutting the microphone noise floor by 16 dB.

PROJECTS

Autonomous Hexapod Spider Robot

Summer 2026

Course Project, EECE 2160 — C++/OOP, Cyclone V SoC (DE10-Nano), Embedded Linux, Verilog

- Programmed an 18-servo hexapod in C++, driving FPGA PWM over memory-mapped I/O to walk with a tripod gait.
- Added autonomous obstacle avoidance: ultrasonic sonar as a Verilog FPGA peripheral and a stop-reverse-turn loop at 20 cm.

Traffic Simulator, Network Routing Engine & Biomedical Signal Processing

Spring 2026

Course Projects, EECE 2140 / EECE 2150 (Team of 3) — C++, OOP, Smart Pointers, MATLAB, SPICE, Git/GitHub

- Built a discrete-time C++ intersection simulator across 7 classes: queuing, light synchronization, emergency override.
- Built a C++ routing engine finding bandwidth-feasible paths (DFS) with leak-free smart pointers (Valgrind-verified); characterized op-amp filters in SPICE and processed real ECG in MATLAB with the FFT to detect heartbeats and remove 60 Hz noise.

Healthy Habit Adventure — Interactive Health-Education Exhibit

Fall 2025

Cornerstone of Engineering (GE 1501), Team of 4 — Raspberry Pi Pico, MicroPython, SolidWorks, 3D Printing

- Proposed the concept the team ranked first in the design matrix — a museum exhibit teaching children healthy habits via a Raspberry Pi Pico quiz game with servo-driven cylinders and NeoPixel feedback; designed the 3D-printed cylinders in SolidWorks.

LEADERSHIP & COMMUNITY

Roxbury Robotics Club, Boston, MA

Sep 2025 – Dec 2025

STEM Instructor (Volunteer)

- Taught mathematics, physics, and engineering fundamentals to children through hands-on LEGO robotics projects.

SKILLS

Programming & Tools: C++, Python (numpy/scipy), MATLAB, R / RStudio, C, MicroPython, Verilog, Bash; Quartus Prime, PSpice / LTspice, Fusion 360, SolidWorks, AutoCAD, 3D printing; Git & GitHub, Linux / WSL, LaTeX

Electronics & Instrumentation: Op-amp, diode, BJT and MOSFET amplifier design; CMOS logic; oscilloscope, function generator, digital multimeter, breadboarding, soldering

Embedded & DSP: RK3588S & Cyclone V SoC bring-up, FPGA and memory-mapped I/O, embedded Linux, ALSA / PortAudio, real-time audio callbacks, adaptive filtering (Kalman / NLMS), FFT & WOLA, FIR / IIR design, Z-transform & DTFT, sampling and quantization

Modeling & Automation: Probability and random processes, regression, hypothesis testing, Monte Carlo simulation (R); PLC ladder logic (TIA Portal), offline robot programming (OCTOPUZ), CNC/CAM

Languages & Interests: Mandarin Chinese (Native), English (Professional working proficiency); quantitative trading, cycling